

2019 Sales Outlook & Demand Intelligence Summary

Prepared for the Head of Supply Chain and the CFO • Based on 2015–2018 sales history

Executive Summary

Over the past four years, sales have grown steadily, driven mainly by the Technology product category, and the business follows a predictable seasonal pattern: demand peaks around the November–December holiday season and drops sharply in January and February. Looking ahead to the first quarter of 2019, we expect a typical seasonal dip in January and February followed by a strong rebound in March, with Technology products leading growth. A handful of unusual sales weeks were identified and are most likely explained by this same holiday-driven seasonality rather than data problems. Products fall into four demand groups that call for different stocking strategies, and we outline three concrete actions below to protect cash flow and inventory efficiency going into next quarter.

Key Findings

- **Technology leads revenue:** Technology generated \$827,456 over the period, ahead of Furniture (\$728,659) and Office Supplies (\$705,422).
- **The East region grows most reliably:** East had the steadiest year-over-year growth of any region, with far less swing than Central, South, or West, making it the lowest-risk region for further investment.
- **Strong, repeatable seasonality:** November and December average roughly three to six times the sales of January and February each year, and the overall trend has been upward since 2015.
- **Shipping is consistent nationwide:** Average delivery time is about 4 days and barely varies by region, so shipping speed is not a current differentiator or risk.

2019 Sales Forecast

We tested three forecasting approaches (SARIMA, Prophet, and XGBoost) and validated each against three months of actual, unseen sales history. The Prophet model was the most accurate of the three and is the basis for the figures below. “Likely Low” and “Likely High” reflect a realistic range around the expected number, based on the model's historical accuracy — actual results are expected to land inside this range in most months.

Month	Expected Sales	Likely Low	Likely High	vs. Dec 2018
January 2019	\$42,100	\$36,000	\$48,300	Seasonal dip
February 2019	\$30,300	\$25,900	\$34,700	Lowest month
March 2019	\$84,300	\$72,000	\$96,600	Sharp rebound

By category, Technology is projected to grow roughly 50% by March versus December 2018, while Furniture and Office Supplies are projected to decline. By region, East is projected to keep growing modestly (+4%) while West is roughly flat. These category and region swings should be treated as directional signals for planning, not guaranteed outcomes (see Risk section).

Top Anomalies Detected

Two independent detection methods were used: one that flags globally extreme weeks (Isolation Forest, 11 weeks flagged) and one that flags sudden local shifts versus the recent trend (Rolling Z-Score, 6 weeks flagged). Only one week was flagged by both.

- **1. Week of March 22, 2015** — the only week both detection methods agreed was abnormal. Likely cause: an unusually large single order or a batch of orders early in the business's history, when overall volumes were low enough that one event could swing the weekly total.
- **2. Holiday-season peak weeks (Nov–Dec)** — flagged as extreme highs. Likely cause: seasonal promotions and year-end purchasing, consistent with the strong November/December pattern seen every year in the data.
- **3. Post-holiday trough weeks (Jan–Feb)** — flagged as extreme lows. Likely cause: the predictable post-holiday slowdown that follows the Nov–Dec surge each year.

Product Demand Segments & Stocking Strategy

Sub-categories were grouped into four demand profiles based on sales volume, growth, volatility, and average order size.

Segment	Example Products	Recommended Stocking Strategy
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High Volume, Stable	Binders, Chairs, Phones, Storage, Tables	Standard replenishment cycles; negotiate bulk-pricing contracts with suppliers.
Low Volume, Volatile	Copiers, Machines	Just-in-time ordering; keep only narrow safety stock to limit cash tied up in slow, expensive items.
Growing Demand	Accessories, Appliances, Art, Labels, Paper	Raise safety stock and build buffer inventory ahead of seasonal spikes.
Declining Demand	Bookcases, Envelopes, Fasteners, Furnishings, Supplies	Run clearance discounts, cut bulk orders, and phase out weak variants.

Business Recommendations

- **1. Shift working capital and marketing toward Technology ahead of Q1.** It is already the top-revenue category and is forecast to grow ~50% by March 2019 — the strongest movement of any segment in the data.
- **2. Prioritize the East region for expansion investment.** East combines the steadiest historical growth of any region with a positive Q1 2019 forecast, making it the lowest-risk region to back with additional supply chain or marketing spend.
- **3. Build a seasonal playbook around the Nov–Dec surge and Jan–Feb trough.** Pre-position inventory and staffing for “Growing Demand” and “High Volume, Stable” segments before the holiday peak, and shift to just-in-time ordering and clearance pricing for “Declining Demand” items heading into the January–February slowdown.

Risk & Limitation

This system is built on four years of internal sales data only — it does not account for external factors such as competitor actions, promotions, pricing changes, or the broader economy, and it has just 48 monthly data points to learn from, which is a thin base for a statistical model. Category-level forecasts (e.g., the projected 55% drop in Furniture) can be more volatile than the overall company forecast and should be sanity-checked against business context before driving purchasing decisions. We also found that the model comparison numbers favored Prophet as the most accurate model, while the analysis narrative initially pointed to SARIMA; this report uses the more accurate model, but it underscores the need for an automated, metrics-based model selection step before any forecast is used operationally.